

Asmaa Abdul-Amin

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Summary

Clearance: Active Secret

Analytical and mission-focused engineer specializing in **web development, software systems, automation, and data integration**.

Experienced with supporting **aerospace and defense missions** at the Johns Hopkins University Applied Physics Laboratory (APL). Interested in advancing **mission-critical** software systems through **secure, scalable engineering solutions**.

Experience

Computer Science/Computer Engineering Intern – Pathways Program

The Johns Hopkins Applied Physics Laboratory – Space Exploration Sector | May 2025 – Present

- Continuing to fortify and **optimize Space Sector web infrastructure**, boosting reliability by ~30% and ensuring compliance across 15+ NASA mission platforms.
- Automated Python + SQL workflows**, cutting data processing time by 40% and saving analysts 10+ hours weekly.
- Partnered with engineering, compliance, and communications teams to **deliver scalable systems with zero critical security findings**.
- Collaborated with cross-functional teams and **presented findings to stakeholders**, ensuring data-driven solutions aligned with mission and business needs

Software & Web Infrastructure Intern – ATLAS Program

The Johns Hopkins Applied Physics Lab – Space Exploration Sector | May 2024 – May 2025

- Developed **Python automated ETL pipelines to automate file format conversion**, strengthening **requirements traceability** and **system interoperability** across mission platforms supporting 100+ engineers, demonstrating skills in **data ingestion, transformation, and orchestration** for high-frequency environments.
- Continued to support **web infrastructure modernization** efforts across multiple mission platforms, ensuring secure, scalable, and compliant backend operations.

Undergraduate Research Intern – CIRCUIT Program

Johns Hopkins APL – Research & Exploratory Development Department | Mar 2023 – May 2024

- Engineered and **deployed multiple webpages for the NASA Dragonfly** internal portal, streamlining access to critical mission resources and improving collaboration efficiency.
 - Built ML model analyzing COVID-19 policies vs. mortality**, improving interpretability of public health data.
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Projects

- AlphaGoat – GAN-based market simulator** with a **reinforcement-learning trader** that generates realistic 1-minute stock data and trains a DQN agent for long/flat/short decisions on live Alpaca feeds.
 - Algo-bot Trading Assistant** – Real-time trading system using **Flask, WebSockets**, and the **Alpaca API** for live portfolio analytics and execution.
 - CryoET Object Identification** – Trained **PyTorch deep learning models** for 3D protein detection, improving recall in high-dimensional biomedical datasets.
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Education

A.A. Computer Science – Montgomery College (Expected Dec 2025)

Focus: Data Analysis, Machine Learning, and Algorithmic Systems, Python Programming

Certifications:

- The Johns Hopkins University Applied Physics Laboratory – Python Programming II (2025)
 - Univ. of Chicago – *Quantum Computing Systems Design I* (2024)
 - IBM – *Machine Learning with Python* (2023)
 - Harvard – *CS50 Intro to Computer Science* (2022)
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Skills

- Languages:** Python (pandas, NumPy, scikit-learn), SQL, C++, Java, HTML/CSS, JavaScript
 - Libraries & Tools:** Flask, REST APIs, Docker, Git, Linux, Jupyter, Kubernetes, Terraform
 - Analytical & ML Tools:** Alpaca API, Backtrader, yfinance, Qiskit, PyTorch, TensorFlow
 - Data & Automation:** ETL pipelines, CI/CD, HPC/cluster computing, statistical analysis
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Additional

- Languages: Arabic (C1 – Advanced)